

Cell Design Specification - Smartphone Battery

Case ID: VBF-T6R1MIMO-DS-001

Date: 2026-08-31

Status: Design Not Achieved (Thermal Constraint Incompatible)

1. Basic Specification

Parameter	Value	Source
Electrochemical System	NMC811 / Graphite	Chen2020 parameter set
Nominal Capacity	4.96 Ah	baseline_1c_discharge.json:capacity_ah
Voltage Window	2.5 - 4.2 V	Chen2020 parameter set
Cell Dimensions	Not provided (shell thickness not in parameter set)	-
Electrolyte Formulation	EC/EMC + LiPF6 (baseline)	Chen2020 default
Electrolyte Additive	High-conductivity formulation ($\sigma=1.8$ S/m, $D=6e-10$ m ² /s, $t_+=0.45$)	elec
Cation Transference Number	0.45	params_elec_e.json

2. Electrode and Separator

Layer	Thickness (μm)	Porosity	Material
Positive Electrode	63	0.32	NMC811
Positive Current Collector	16	-	Aluminum
Separator	25	0.47	Polyethylene
Negative Current Collector	10	-	Copper
Negative Electrode	73.3	0.32	Graphite

N/P Ratio: 1.05 (negative capacity / positive capacity)

3. Process Design Parameters

Parameter	Value	Unit	Formula/Source
Positive Areal Density	10.6	g/m ²	thickness × (1 – porosity) × density
Negative Areal Density	12.8	g/m ²	thickness × (1 – porosity) × density
Positive Compaction Density	3.2	g/cm ³	electrode density × (1 – porosity) / 1000
Negative Compaction Density	1.8	g/cm ³	electrode density × (1 – porosity) / 1000
Electrolyte Fill Amount	Not provided	mL	electrolyte density not in parameter set
Formation Recommendation	0.1C CC to 4.2V, 25°C, 2 cycles	-	Design recommended value

4. Mass Breakdown

Layer	Mass (g)	Source
Positive Electrode	16.4	baseline_energy.json:layer_kg_m2.positive_electrode × area
Negative Electrode	10.6	baseline_energy.json:layer_kg_m2.negative_electrode × area
Positive Current Collector	4.3	baseline_energy.json:layer_kg_m2.positive_cc × area
Negative Current Collector	10.8	baseline_energy.json:layer_kg_m2.negative_cc × area
Separator	0.25	baseline_energy.json:layer_kg_m2.separator × area
Total Cell Mass	**43.5**	baseline_energy.json:mass_kg × 1000

Note: Electrolyte mass not included (parameter set lacks electrolyte density)

5. Performance Verification Table

Metric	Target	Achieved	Pass/Fail	Source
Volumetric Energy Density	≥ 950 Wh/L	852.8 Wh/L	✗	elec_e_energy.json:energy_density_wh_l
Gravimetric Energy Density	-	404.7 Wh/kg	-	elec_e_energy.json:energy_density_wh_kg
Voltage Plateau	≥ 4.1 V	3.97 V	✗	elec_e_energy.json:midpoint_voltage_v
4C Fast Charge (No Plating)	No plating	No plating	✓	elec_e_4c_charge.json:anode_potential_
Maximum Temperature (4C)	$\leq 50^{\circ}\text{C}$	79°C	✗	elec_e_4c_charge.json:T_max_K
SEI Thickness (100 cycles)	≤ 500 nm	482 nm	✓	elec_e_aging_100cyc.json:sei_thickness_nm_end
DC Resistance	-	0.148 m Ω	-	elec_e_energy.json:dcr_ohm
Power Density	-	634.6 kW/kg	-	elec_e_energy.json:power_density_w_kg

6. Design Notes

What Was Changed

- **Electrolyte Transport Properties****: Increased conductivity from baseline ~ 1.0 S/m to 1.8 S/m, d
- **Rationale****: High-transport electrolyte eliminates lithium plating during 4C fast charge (anode

Why Changes Were Made

- Baseline Chen2020 system showed lithium plating during 4C charge (anode potential -0.44V)
- Three architecture variants (thick/thin/porosity) all failed to eliminate plating
- Electrolyte transport improvement was identified as the only viable path to eliminate plating

What Could Not Be Achieved

- **Thermal Constraint (50°C at 45°C ambient)****: Physically incompatible with 4C fast charge. Minim
- **Energy Density (950 Wh/L)****: Chen2020 system ceiling is ~ 850 Wh/L. Would require high-capacity
- **Voltage Plateau (4.1V)****: Chen2020 NMC811 peaks at ~ 4.04 V. Would require higher-voltage cathode

Recommendations for Future Design

1. Relax thermal constraint to 80°C maximum for 4C charge at 45°C ambient
2. Use OKane2022 parameter set with SiOx anode for higher energy density
3. Consider LNMO high-voltage cathode for 4.1V+ plateau